

Literature Status: The Factorial Group-Algebra Completion

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Status. literature_already_answered. A later primary paper by Cahen–Gutt–Waldmann [2] explicitly identifies and analyzes the group-algebra completion left open by Beiser–Waldmann [1].

Original questions. For a finitely generated infinite group G with word length L , the source rescales the group basis by

$$u_g = \frac{g}{(L(g)!)^\epsilon} \quad (\epsilon > 0)$$

and defines a Fréchet topology by recursive seminorms. Remark 3.14 asks how the topology depends on the generators and on ϵ , whether the completion is exactly the formal series with subfactorial coefficient growth, and for a useful tensor topology making the coproduct continuous.

Exact later identification. Section 5 of [2] states that understanding this recursion was one motivation for the later paper and reconstructs it with $c(g) = (L(g)!)^\rho$. Lemma 5.1 proves estimates in both directions, and Proposition 5.2 proves equality of Fréchet algebras and of their locally convex topologies:

$$\mathcal{A}_{L,\rho}(G) = \ell_{L,n!,\rho-}^1(G). \quad (1)$$

The right side consists of canonical-basis series $x = \sum_g x_g g$ such that

$$\sum_{g \in G} |x_g| (L(g)!)^R < \infty \quad \text{for every } R < \rho. \quad (2)$$

This is precisely the requested subfactorial description in the source’s rescaled coordinates. Indeed, if $x = \sum_g b_g u_g$, then $x_g = b_g / (L(g)!)^\rho$, and (2) becomes

$$\sum_{g \in G} \frac{|b_g|}{(L(g)!)^\delta} < \infty \quad \text{for every } \delta > 0, \quad (3)$$

where it suffices to take $0 < \delta < \rho$. Thus the completion consists exactly of subfactorially growing rescaled coefficients, with the corresponding Köthe topology.

Dependence on generators and parameter. Immediately after Proposition 5.2, the later paper states that $\mathcal{A}_{L,\rho}(G)$ generally depends on the word length L ; independence is obtained only after taking the projective limit $\rho \rightarrow \infty$. Formula (2) also makes the parameter dependence explicit: increasing ρ requires faster factorial decay. For an infinite group these spaces are strictly nested, as seen on a geodesic ray $L(g_n) = n$ by taking $x_{g_n} = (n!)^{-\alpha}$ and all other coefficients zero.

Coproduct. For the weighted norm $\|x\|_{L,\sigma,R} = \sum_g |x_g| \sigma(L(g))^R$, equation (4.7) of [2] gives the exact relation

$$\|\Delta x\|_{R\widehat{\otimes}_\pi R} = \|x\|_{2R}. \quad (4)$$

This explains the fixed-window obstruction and the remedy: take the projective limit over all $R \geq 0$, so every target seminorm has the needed $2R$ source seminorm. Proposition 4.4 and Theorem 4.5 then give a length-independent Fréchet–Hopf $*$ -algebra, functorial in finitely generated groups.

Identification and scope. This is a direct later-paper answer, not a same-paper ask-and-answer passage. The later paper cites [1], reproduces its recursion, proves the exact topological identification, and addresses the length and coproduct issues. No new mathematical solution or priority claim is made here.

References

- [1] S. Beiser and S. Waldmann, *Fréchet algebraic deformation quantization of the Poincaré disk*, J. Reine Angew. Math. **688** (2014), 147–207; arXiv:1108.2004v3.
- [2] M. Cahen, S. Gutt, and S. Waldmann, *Nuclear Group Algebras for Finitely Generated Groups*, Bull. Belg. Math. Soc. Simon Stevin **27** (2020), 567–594; arXiv:1610.07746.